



**UMPSA
TECHNOLOGY**
(Formerly known as UMP Technology Sdn. Bhd.)

Universitiy Malaysia Pahang Al-Sultan Abdullah
UMPSA Technology

26600, Pekan, Pahang
Malaysia
Phone. +609 431 5047

Company / customer	JABATAN PENGURUSAN PENGETAHUAN DAN TEKNOLOGI MAKLUMAT
Project description	INDUSTRIAL POWER MONITORING SYSTEM
Job number	PR000001
Project Type	Energy Monitoring System

Manufacturer (company)

Project name	Energy_Substation
Installation Year	2025
Place of installation	Substation Power UMPSA
Responsible for project	AmirulA

Created on 20/12/2024

Edit date 17/12/2025

by (short name) Alinuh

Number of pages 12

SAFETY REGULATIONS

Universiti Malaysia Pahang Al-Sultan Abdullah
UMPSA Technology
26600, Pekan, Pahang
Malaysia



GENERAL SAFETY REGULATIONS



DANGER !

During plant operation certain items are under dangerous voltage!
Non-observance of safety instructions can result in death, serious injuries or material damage.

Only specialist personnel may carry out transport, installation and commissioning work.

The applicable valid standards as well as the national and / or company-specific accident prevention regulations are to be observed.

The following safety instructions are to be observed:

The erection, commissioning, troubleshooting as well as repair of the plant may only be carried out by qualified personnel that is familiar with the corresponding operating instructions.

Mounting of the devices has to be effected in accordance with the valid standards, state and local regulations.
Proper grounding and conductor dimensioning as well as proper short-circuit proofing have to be ensured.
These measures serve to ensure the safety of the plant and of the operating personnel.

Before carrying out safety checks, maintenance work and repair measures ensure that all the power supplies are switched off, are secured against being switched on unintentionally and are marked correspondingly.

Only test devices that are in a technically perfect condition and are suitable for the respective measurement may be used to carry out measurements!

The instructions specified in the respective operating instructions are to be followed strictly!
It is mandatory that hazard, warning and safety instructions be observed!

All doors and covers are to be kept closed during the plant operation.
If cooling devices are installed in the plant, ensure that these systems operate trouble-free.
These include the regular cleaning of the filters, in as far as they exist.

Plant designation INDUSTRIAL POWER MONITORING SYSTEM						Drawing number 001	
Customer Name						Customer Name	
Company						Company	
Ed. 28/12/2024 EKAN						University Malaysia Pahang Al-Sultan Abdullah	
Appr. 17/12/2025 AINAH						UMPSA Technology	
						26600, Pekan, Pahang	
						Malaysia	
Safety Regulation						Higher-level function	
Original						Mounting Location	
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DCB		1	Title page		17/12/2025	Alinuh
		2	Safety Regulation		17/12/2025	Alinuh
		3	Table of Content		17/12/2025	Alinuh
		1	Cover Sheet		9/8/2023	Alinuh
	A1	1	Main Control Panel AC Input 230 VAC 50 Hz		27/8/2025	yatih
	A1	2	24 VDC device power supply		17/12/2025	Alinuh
	A1	3	Input Diagram		17/12/2025	Alinuh
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	A1	6	ENCLOSURE 400x300x200 Overall View		17/12/2025	Alinuh
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	A1	1	Summarized Part list		17/12/2025	Alinuh
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Plant designation INDUSTRIAL POWER MONITORING SYSTEM						Drawing number F06_UMP_001		 UMPSA TECHNOLOGY UNIVERSITY MALAYSIA PAHANG	
Customer Name Company University Malaysia Pahang Al-Sultan Abdullah UMPSA Technology 25600, Pekan, Pahang Malaysia									
Modification	Date	Name	Ed.	Date	Name	Table of Content			
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			Appr.	17/12/2025	Alinuh				
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COVER SHEET

Universiti Malaysia Pahang Al-Sultan Abdullah
UMPSA Technology
26600, Pekan, Pahang
Malaysia



INDUSTRIAL POWER MONITORING SYSTEM

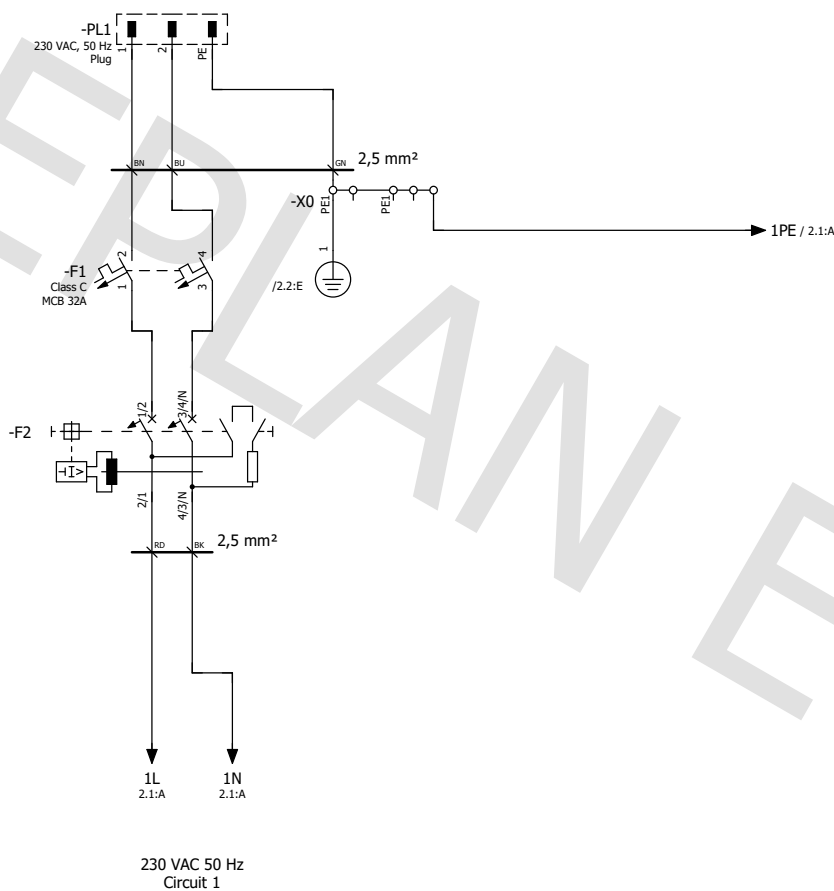
Energy Monitoring System

Work Station:

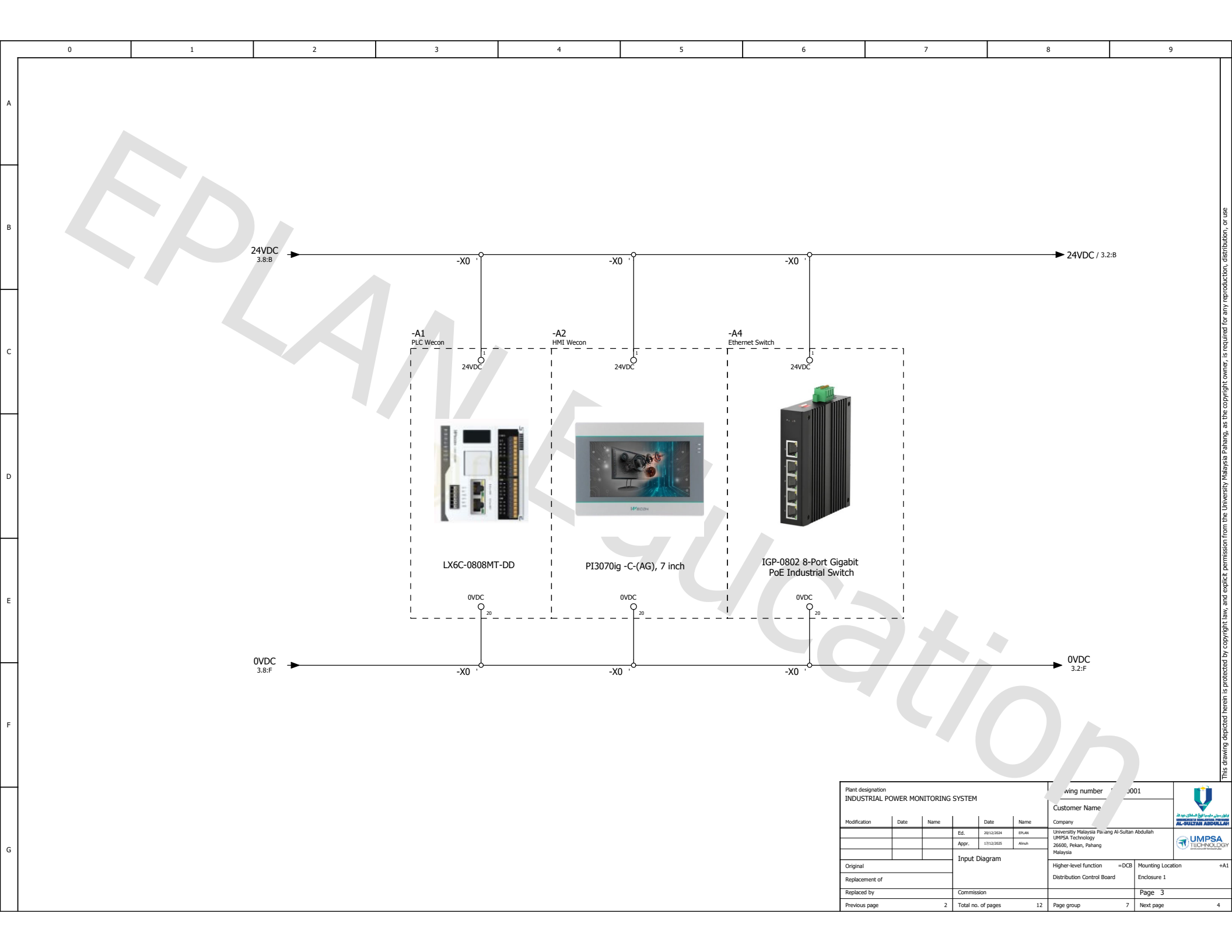
Distribution Control Board

F26_UMPSAT_003

Plant designation INDUSTRIAL POWER MONITORING SYSTEM						Drawing number F26_UMPSAT_003	
Customer Name						Customer Name	
Company						Company	
Ed. 28/12/2024 EKAN						University Malaysia Pahang Al-Sultan Abdullah	
Appr. 17/12/2025 AINAH						UMPSA Technology	
Original						26600, Pekan, Pahang	
Replacement of						Malaysia	
Replaced by						Higher-level function = DCB	
Commission						Mounting Location	
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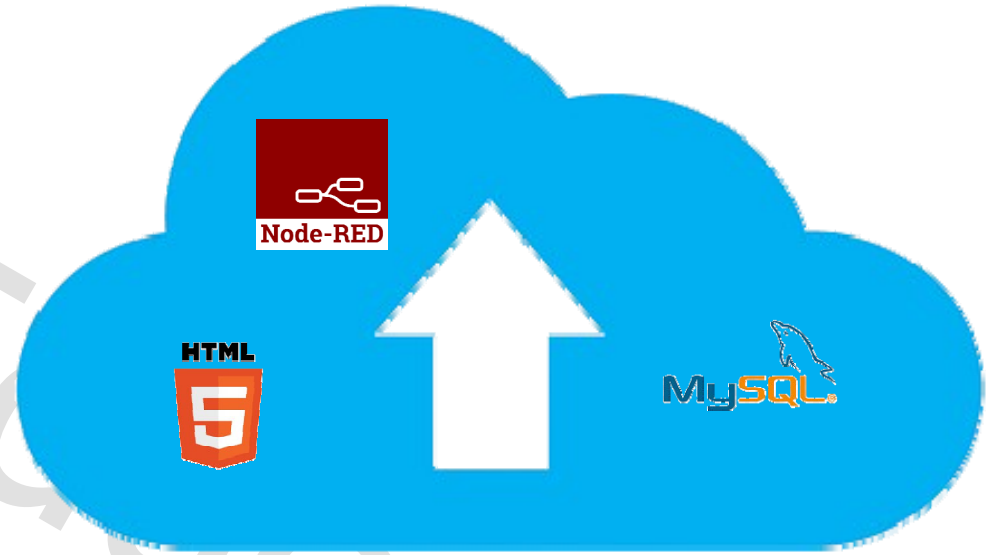
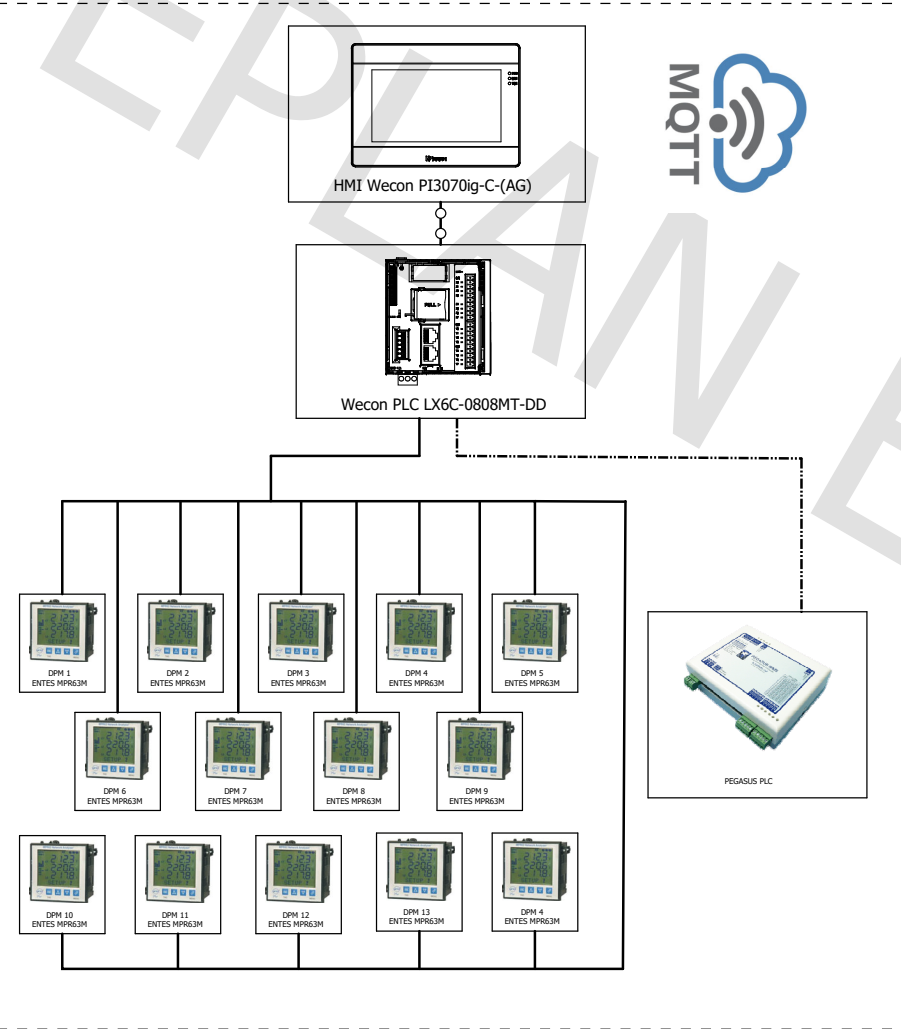
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Customer Name						Company			
Modification	Date	Name	Ed.	Date	Name	University Malaysia Pahang Al-Sultan Abdullah UMPSA Technology 25600, Pekan, Pahang Malaysia			
			Ed.	28/12/2024	IFUAW				
			Appr.	2/06/2025	yeh				
Original			Main Control Panel AC Input 230 VAC 50 Hz			Higher-level function =DCB		Mounting Location =A1	
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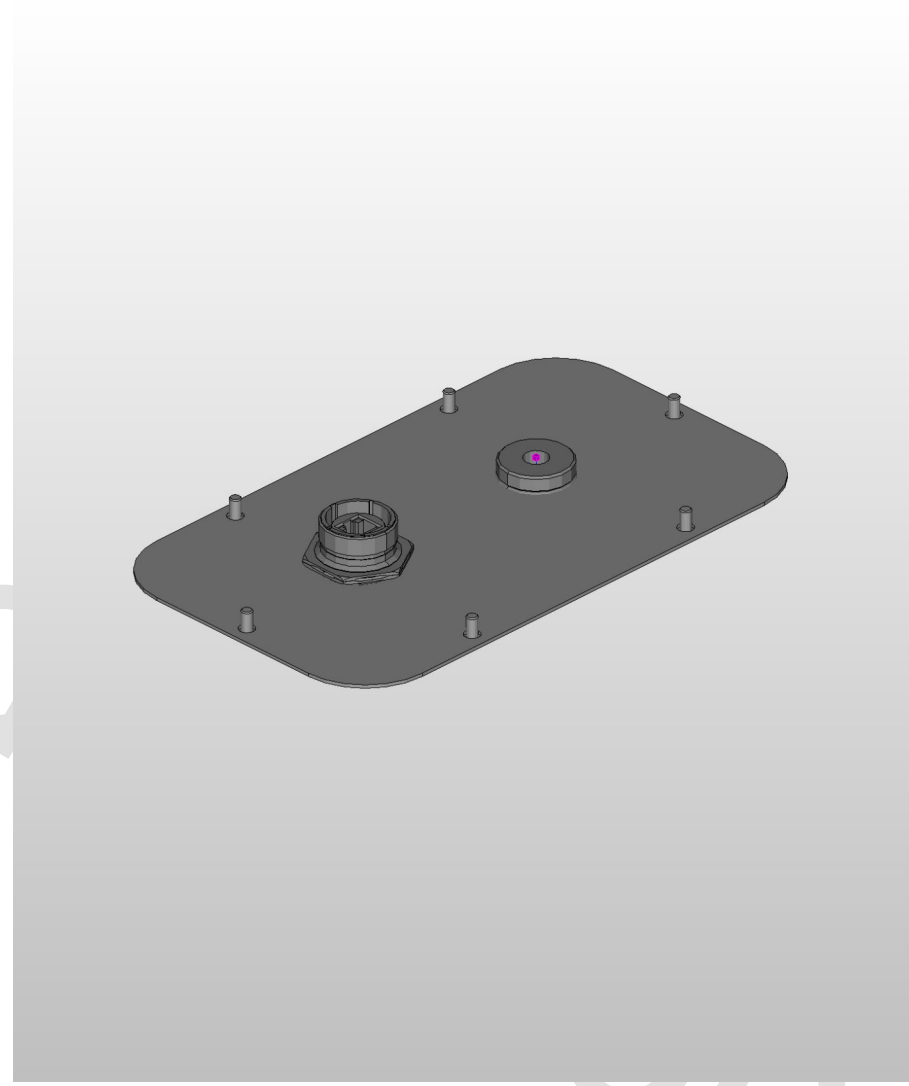
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Customer Name						Company University Malaysia Pahang Al-Sultan Abdullah UMPSA Technology 25600, Pekan, Pahang Malaysia					
Modification										Date	
Name										Name	
Ed.						28/12/2024		EFAN			
Appr.						17/12/2025		Ainuh			
Input Diagram						Higher-level function		=DCB			
Original						Distribution Control Board		Mounting Location			
Replaced by								Enclosure 1			
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LEGEND

- Modbus TCP/IP
- Modbus RTU / RS-485

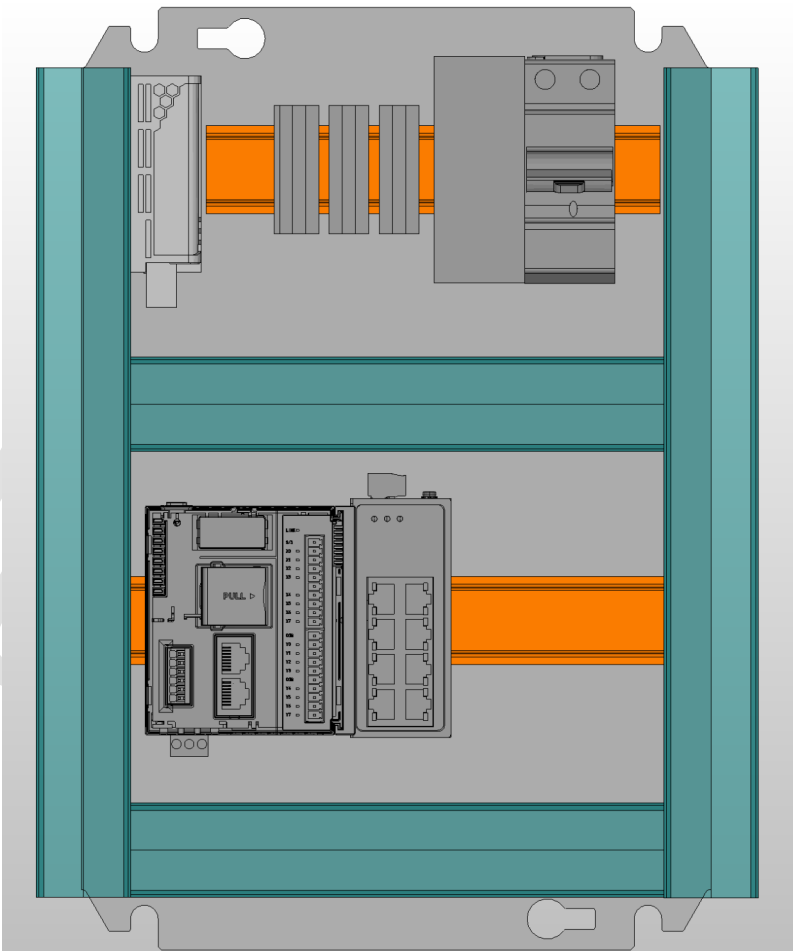
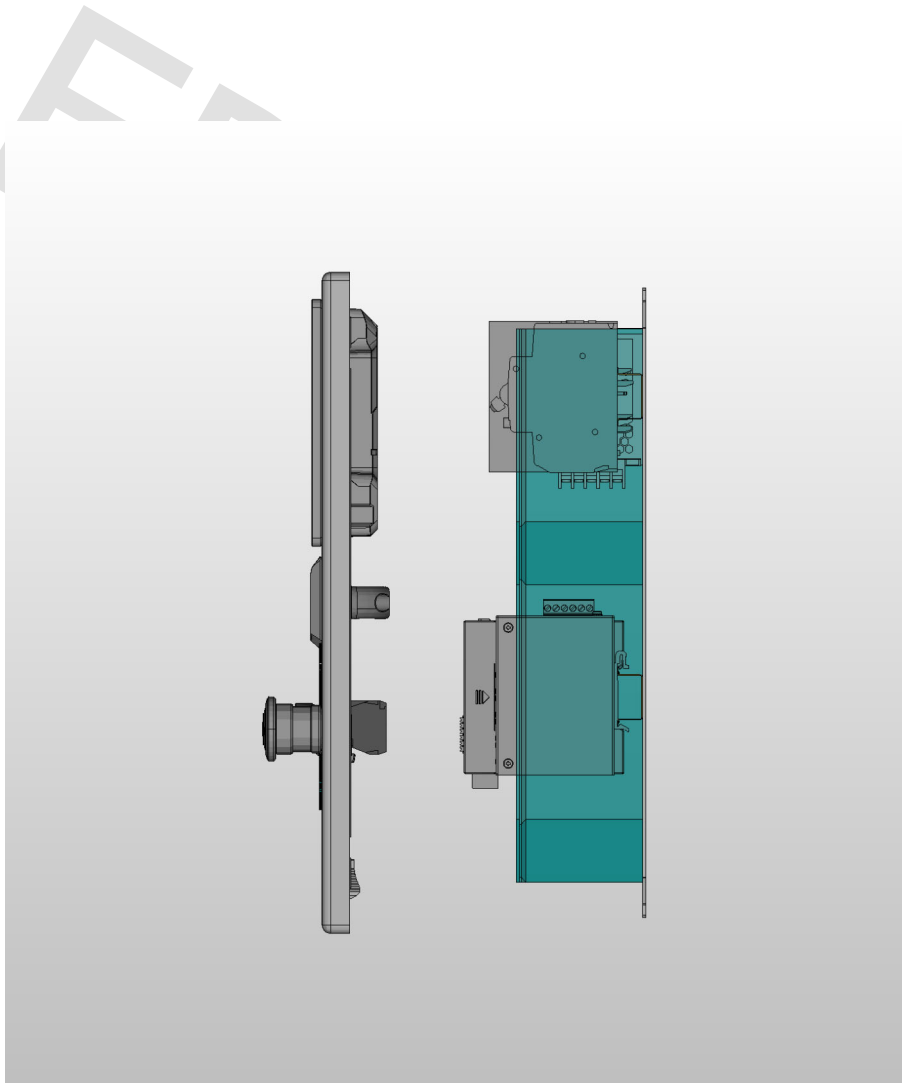


Plant designation INDUSTRIAL POWER MONITORING SYSTEM					Drawing number 001				
Modification					Customer Name				
Date					Company				
Name					University Malaysia Paviang Al-Sultan Abdullah				
Ed.					UMPSA Technology				
Appr.					25600, Pekan, Pahang				
					Malaysia				
Original					Higher-level function = DCB			Mounting Location +A1	
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Plant designation INDUSTRIAL POWER MONITORING SYSTEM					Drawing number 001		 UNIVERSITI MALAYSIA PAHANG AL-SULTAN ABDULLAH UMPSA TECHNOLOGY		
Customer Name					Company				
Modification	Date	Name	Ed.	28/12/2024	ERAN		University Malaysia Pahang Al-Sultan Abdullah UMPSA Technology 26600, Pekan, Pahang Malaysia		
			Appr.	17/12/2025	Alruh				
			ENCLOSURE 400x300x200 Overall View			Higher-level function	=DCB	Mounting Location	+A1
Original						Distribution Control Board			
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Plant designation					Drawing number				
INDUSTRIAL POWER MONITORING SYSTEM					0001				
Modification					Customer Name				
					Company				
					University Malaysia Pahang Al-Sultan Abdullah				
					UMPSA Technology				
					25600, Pekan, Pahang				
					Malaysia				
Original					Higher-level function				
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Ed.		2012/2024		BRLW	
Appr.		1712/2025		Almuh	
ENCLOSURE 400x300x200					
Overall View					
Commission		Total no. of pages		12	
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Summarized parts list

F02_2900_UMPSAT_V0

Position

Quantity

Unit

Part number

Designation

Manufacturer

Order number

Technical characteristics

1

1

LAPP.4150400

Multi-Standard SC 2.2 1X1,5 GNYE

LAPP

4150400

2

1

LAPP.4150402

Multi-Standard SC 2.2 1X1,5 BU

LAPP

4150402

3

2

LAPP.4150401

Multi-Standard SC 2.2 1X1,5 BK

LAPP

4150401

4

1

LAPP.4150406

Multi-Standard SC 2.2 1X1,5 GY

LAPP

4150406

5

1

Pcs

MR0000295

ETHERNET RJ45 GLAND CONNECTOR

UMPSA

6

2

Pcs

MR0000293

Electrical Cabinet Complete

CVS

380x300x200

7

9

Pcs

MR0000037

ALOFREE CABLEDUCT 37.5X75 Black

Canalplast

PA037N

Halogen Free Trunking

8

5

Pcs

MR0000236

RAIL DIN 35

Weidmuller

0236400000

T=1.5mm, L=2mm

9

1

Pcs

MR0000289

Electrical Cabinet Complete

CVS

500x400x200

10

3

pcs

MR0000290

UMPSA LOGO

11

1

Pcs

MR0000294

Electrical Cabinet Complete

CVS

400x300x200

12

1

Pcs

MR0000167

GLAND MS-M32 11-21mm

SKINTOP

12 58 14

M32x1.5

13

1

Pcs

BR1028263

FUSE HOLDER

PXC

3036372

2-CONDUCTOR BLACK

14

1

Pcs

BR79369

FUSE

Littelfuse

33-051-41

Blade fuse

15

3

Pcs

BR1028254

End cover

PXC

3031762

Separator plate for STS 2,5-TWIN

16

18

Pcs

BR1028250

TERMINAL 2,5mm²

Phoenix Contact

3031720

3-CONDUCTOR GRAY

17

9

Pcs

BR1028251

TERMINAL 2,5mm²

PHOENIX

3036275

3-CONDUCTOR BLUE

18

3

Pcs

BR1028253

TERMINAL

PHOENIX

3031733

3-CONDUCTOR PE

19

3

piece

MR0000286

Programmable logic controllers (PLC), Modular PLC, CJ-Series CPU units

WECON

WECON PLC

20

3

piece

MR0000287

WECON HMI 7 INCH

WECON

P13070ig

21

2

piece

MR0000288

IOT GATEWAY

RASPBERRY PI

IRIV PI

22

3

MR0000292

ETHERNET SWTICH

23

3

Pcs

BR1001829

MAIN CIRCUIT BREAKER

SIEMENS

5SY6 210-7

C-CHARACTER

24

3

piece

ABB.2CSF502110R2400

Residual current circuit breaker (RCCB)

ABB

2CSF502110R2400

Residual current operated circuit breaker SMISSLINE,1PN,A,40A,100mA

25

2

Piece

SE.XB4BVBM4GEX

AEx RED LED PILOT LIGHT 24..240 V

SE

XB4BVBM4GEX

26

1

Pcs

BR70972

WIRE RING

Nelco

2538.0

1,5-2,5 mm²

27

3

Pcs

MR0000064

TRIGGER ACTION BUTTON BLACK

TELEMEC

ZB5-AS52

BLACK Ø40 TURN RELEASE

Plant designation
INDUSTRIAL POWER MONITORING SYSTEM

Modification

Date

Name

Ed.

2012/2024

IPUN

Appr.

13/12/2025

Amuh

wing number

0001

Customer Name

Company

University Malaysia Pahang Al-Sultan Abdullah

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26600, Pekan, Pahang

Malaysia

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Summaryzed Part list

Higher-level function

=DCB

Mounting Location

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UMPSA TECHNOLOGY

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B

C

D

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F

G

Summarized parts list

F02_2900_UMPSAT_V0

Position	Quantity	Unit	Part number	Designation	Manufacturer	Order number	Technical characteristics
28	2	Pcs	MR0000040	AUXILIARY BLOCK	TELEMEC	ZB5-AZ105	switch body with 1 NC and 1NO block
29	1	Pcs	MR0000007	Legend Holder	Schieder	ZBZ35	without legend
30	1	Pcs	MR0000006	Legend	KW	1704673	Blank legend
31	3	Pcs	MR0000060	Selector Switch	Schneider Electric	XB4BD21	2-position stay put, complete selector switch
32	1	pcs	MR0000291	Convesional Power Supply	IFM	ZB0805	Output current 10 A, regulated
33	1		TW 004	Temperature measurement with resistance thermometer 1- and 2-fold	ESS		
34	13		LW 001	Measuring set-up for filling level: measurement triggered by float with contact	ESS		
35	10						
36	20		not yet known				

Plant designation
INDUSTRIAL POWER MONITORING SYSTEM

Drawing number
0001

Customer Name
Company

Ed.
2012/2024
Bilal

Appr.
17/12/2025
Almuh

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Summarized Part list

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Higher-level function
=DCB

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Plant designation INDUSTRIAL POWER MONITORING SYSTEM						Drawing number 001		
Customer Name						Customer Name		
Company						Company		
Ed. 28/12/2024 BY:AN						University Malaysia Pahang Al-Sultan Abdullah UMPSA Technology 25600, Pekan, Pahang Malaysia		
Modification		Date	Name			Date	Name	Summarized Part list
						Appr. 17/12/2025	Ainuh	
Original								
Replacement of								
Replaced by								
Commission						Higher-level function =DCB Mounting Location =A1		
						Distribution Control Board Enclosure 1		
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